



National Radio
Astronomy
Observatory

NAASC
NORTH AMERICAN ALMA SCIENCE CENTER

ALMA Proposal Preparation Workshop

March 25, 2026 @INCT-UDA, Copiapó

1.2. ALMA Science and Capabilities

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The Atacama Large Millimeter/submillimeter Array

Array Operations Site (AOS), Chajnantor plateau, Atacama Desert, Chile



66 antennas



ALMA Photo Gallery: <https://alma-telescope.jp/en/gallerytag/aos/>



Credit: ALMA Observatory

The Atacama Large Millimeter/submillimeter Array

Array Operations Site (AOS), Chajnantor plateau



► High-frequency array

- 66 antennas 12m & 7m
- 35–950 GHz (0.3–8.6 mm)
- 5000 m high, cold dry air reduces PWV amount

► ALMA detects:

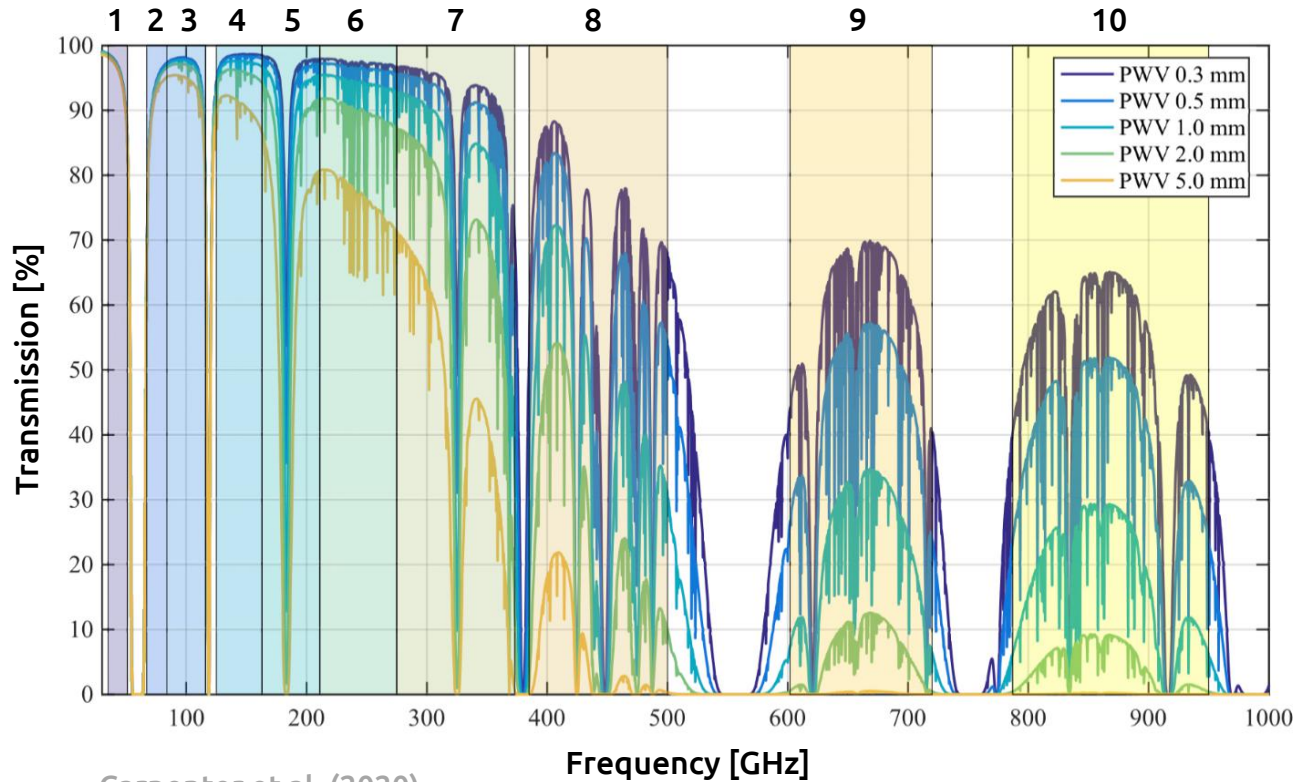
- Thermal dust continuum
- Molecular spectral lines
- Free-free continuum

► Three subarrays

- 12-m Array (50 × 12m)
 - 7-m Array (12 × 7m)
 - Total Power (4 × 12m)
- } ACA

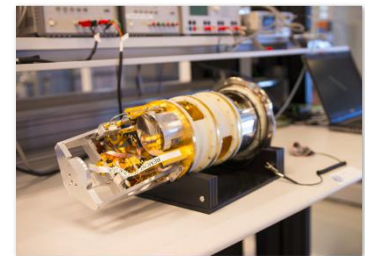
ALMA Receiver Bands

Atmospheric transmission in the ten ALMA receiver bands



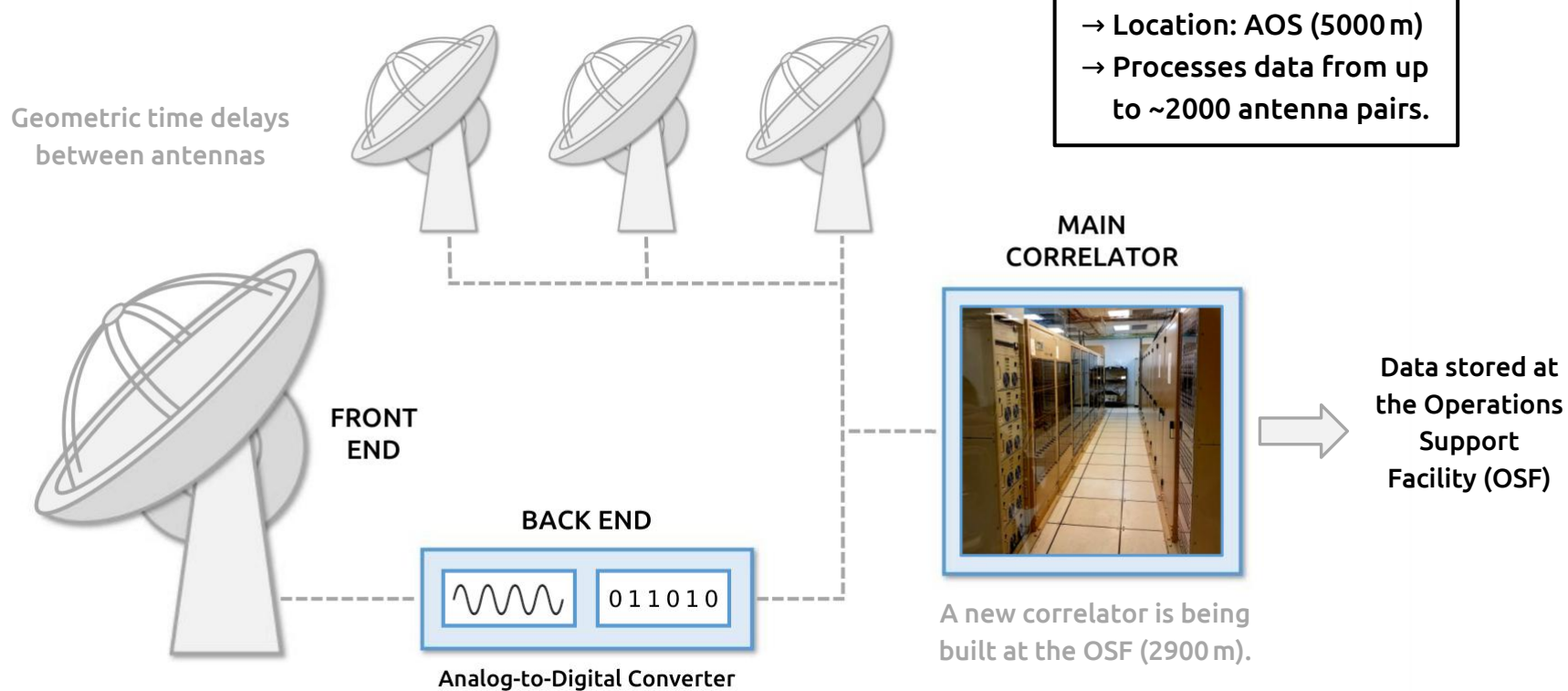
Carpenter et al. (2020)

Receiver cartridges



ALMA Antenna Correlator

front end → digitization → correlation → visibilities



- Sky frequencies $\sim 10^2$ GHz
 - Too high to be processed directly by the electronics.
 - Sky signal is converted to a lower frequency (IF) using a very stable frequency (LO).

IF $\equiv$ Intermediate Frequency

LO $\equiv$ Local Oscillator

- Upper & Lower Sidebands

$$\rightarrow \text{USB} = \text{LO} + \text{IF}$$

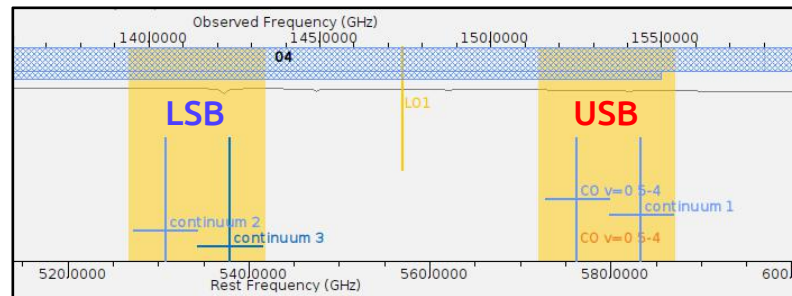
$$\rightarrow \text{LSB} = \text{LO} - \text{IF}$$

Heterodyne principle: receiver bands produce two sidebands.

Note: The lower sideband of Band 1 is filtered out, i.e., this is the only receiver with a single sideband.

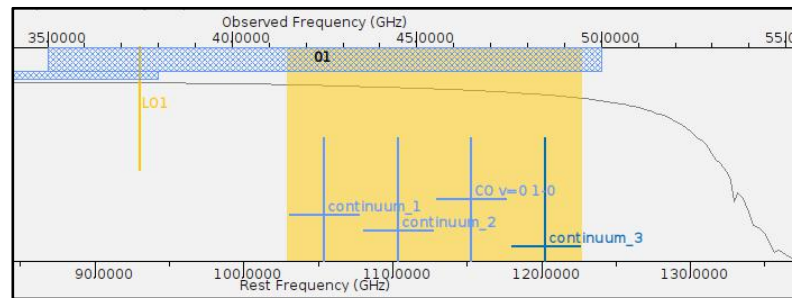
Upper and Lower Sidebands in the ALMA Observing Tool (OT)

Band 4 ➤
(2SB)



Program #2024.1.00015.S, PI Cornil-Baiotto

Band 1 ➤
(SSB)



Program #2025.1.00006.S, PI Cornil-Baiotto

- Sky frequencies $\sim 10^2$ GHz
 - Too high to be processed directly by the electronics.
 - Sky signal is converted to a lower frequency (IF) using a very stable frequency (LO).

IF $\equiv$ Intermediate Frequency

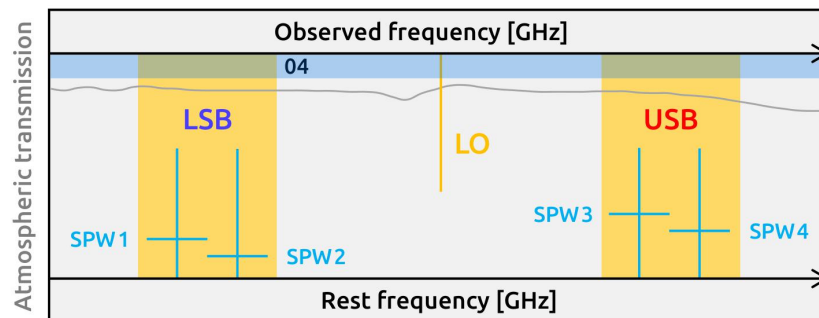
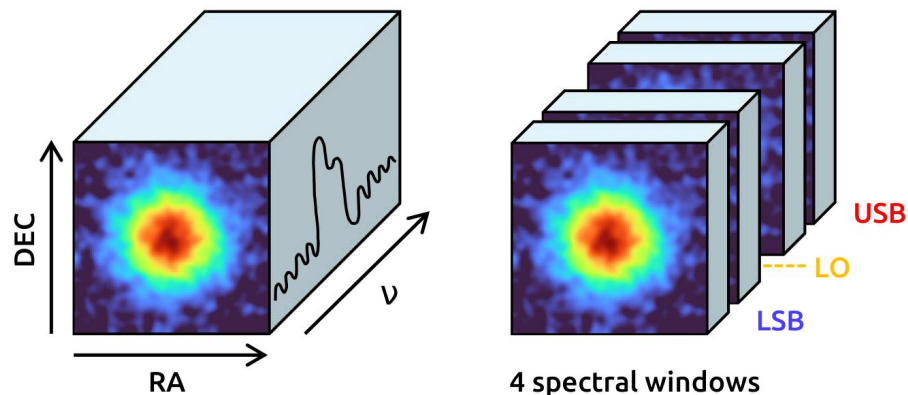
LO $\equiv$ Local Oscillator

- Upper & Lower Sidebands

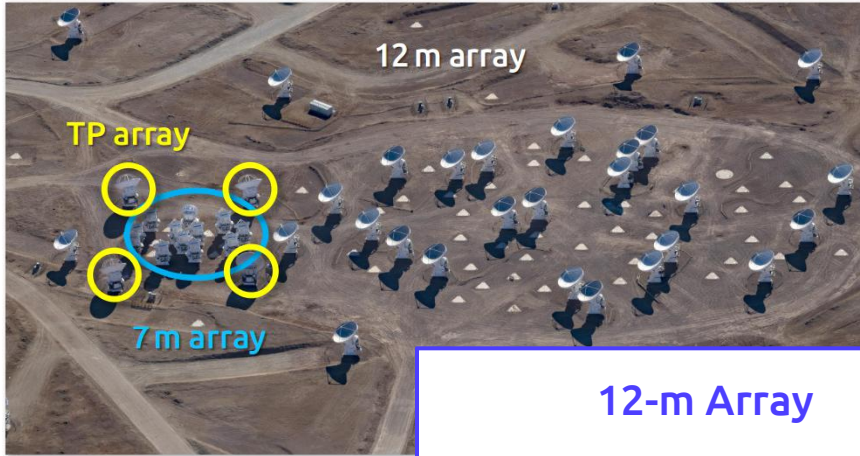
$$\rightarrow \text{USB} = \text{LO} + \text{IF}$$

$$\rightarrow \text{LSB} = \text{LO} - \text{IF}$$

Heterodyne principle: receiver bands produce two sidebands.



ALMA 12m and 7m Array Comparison



12-m Array

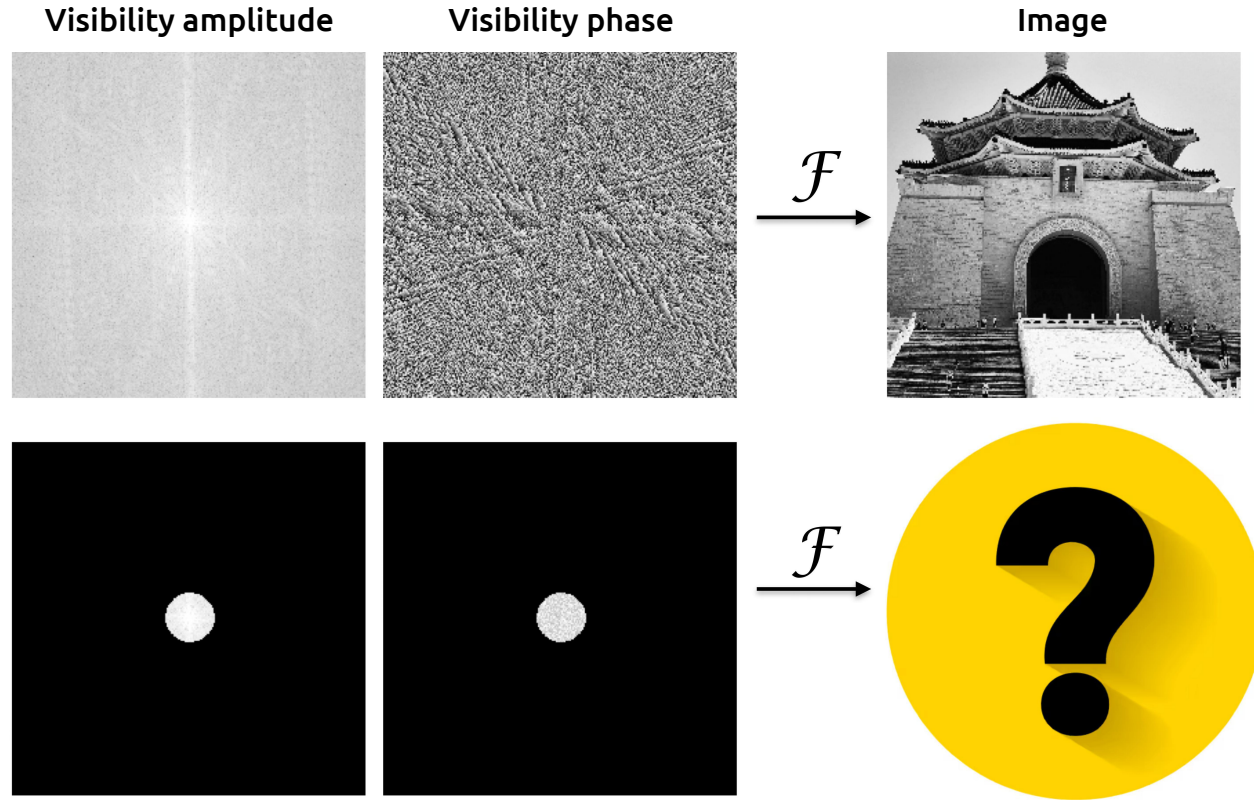
- 50 antennas
- Shortest baseline: 14.6 m
- Longest baseline: 16194 m
- High angular resolution
- Higher sensitivity
- Better UV coverage



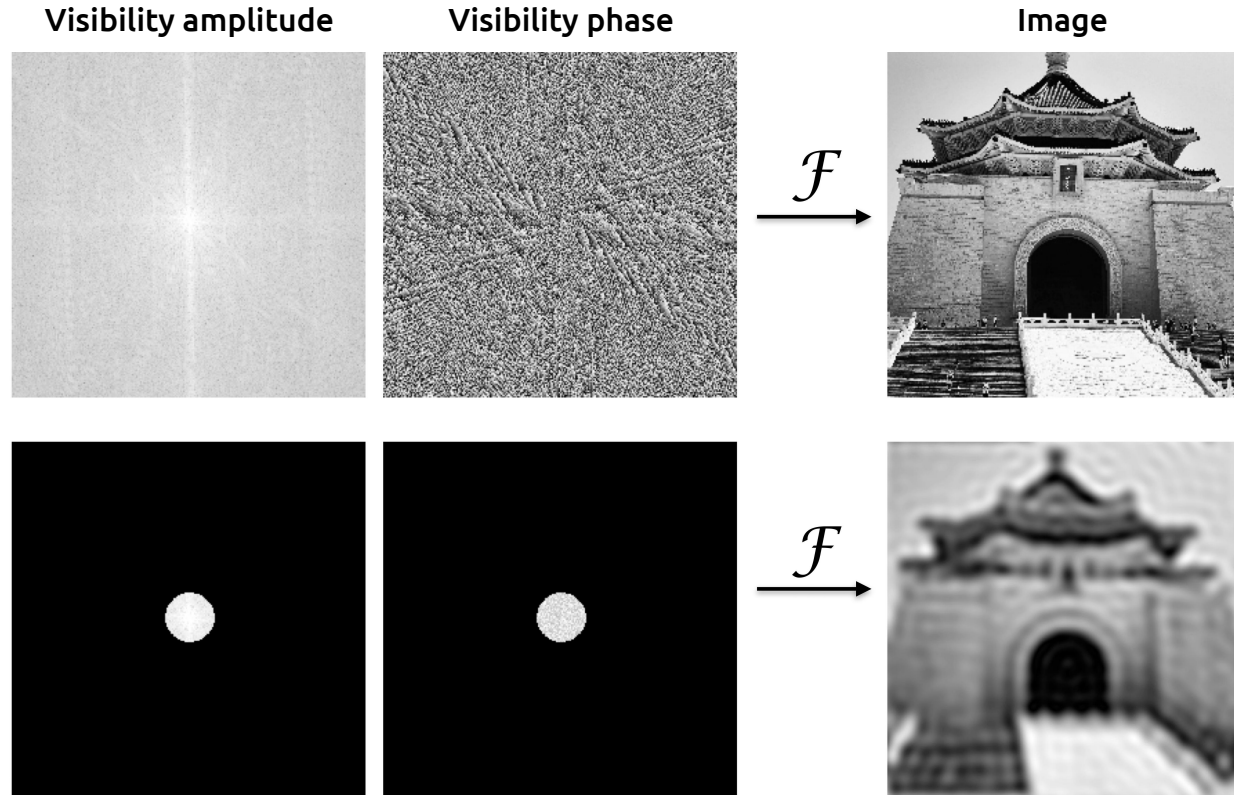
7-m Array

- 12 antennas
- Shortest baseline: 8.7 m
- Longest baseline: 45.0 m
- Sensitive to larger structures
- Lower sensitivity
- Worse UV coverage

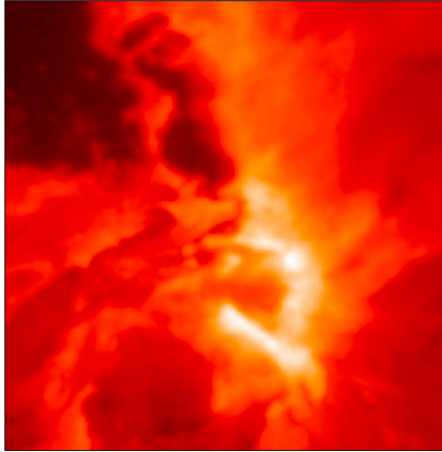
Impact of Finite uv -Sampling



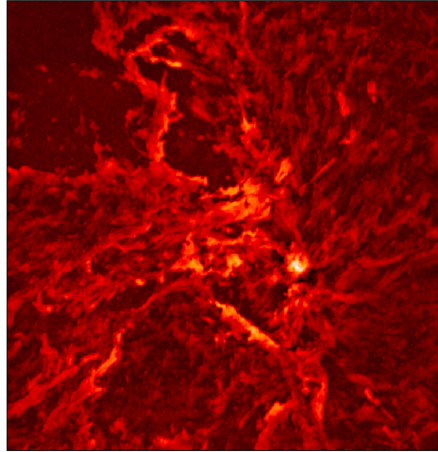
Impact of Finite uv -Sampling



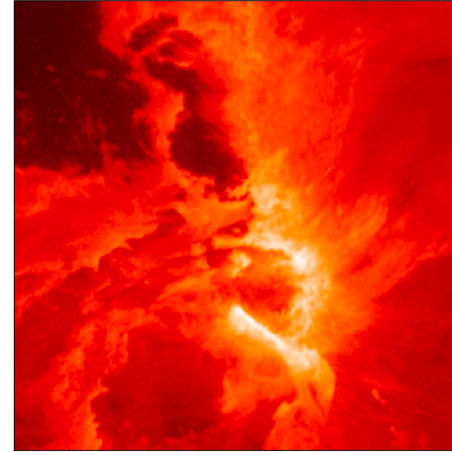
Single Dish



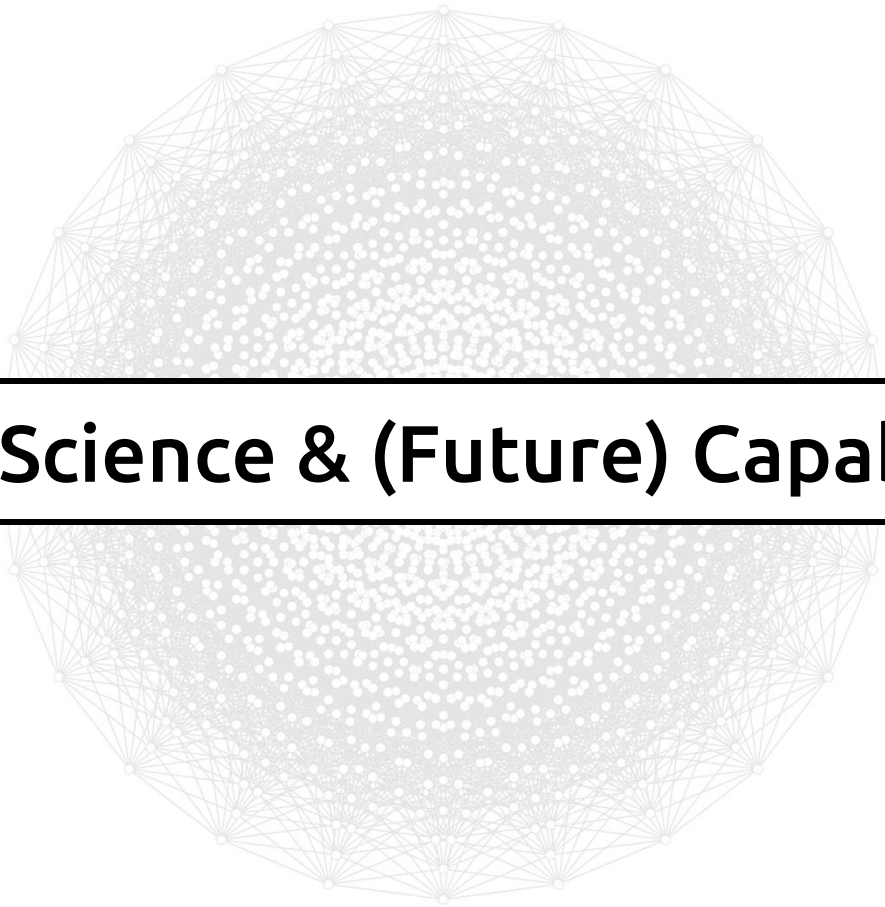
Interferometer



Combined Image

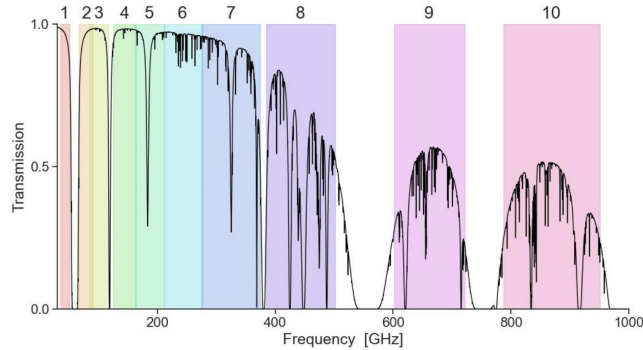


- The 12-m Array probes small spatial scales
- The 7-m Array and single-dish data probe larger scales
- Combining both recovers emission on all spatial scales

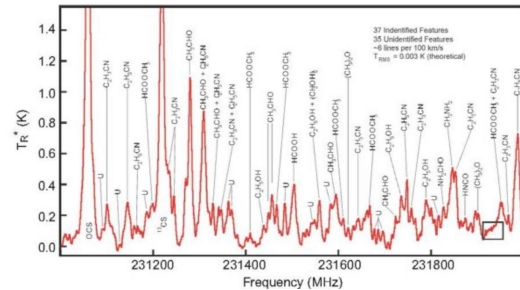


ALMA Science & (Future) Capabilities

(35-950 GHz, $\lambda \approx 0.3 - 9$ mm)



Spectroscopy



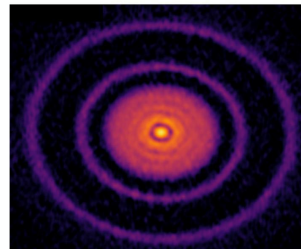
Sensitive spectroscopy to study molecular gas and chemistry

Baselines: 0.16 to 16 km

(resolution as fine as $0.005''$ at 950 GHz)

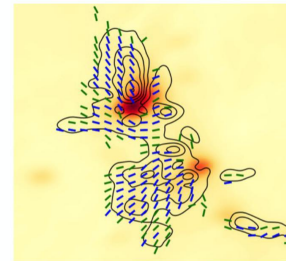


Continuum



Trace dust and SF

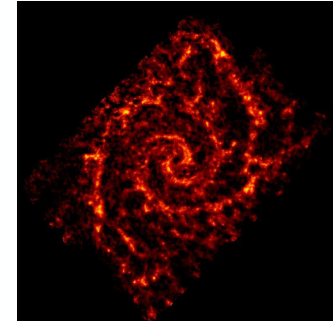
Polarization



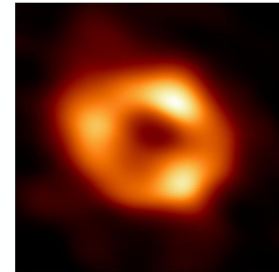
Probe magnetic fields

Imaging

High-fidelity imaging



VLBI

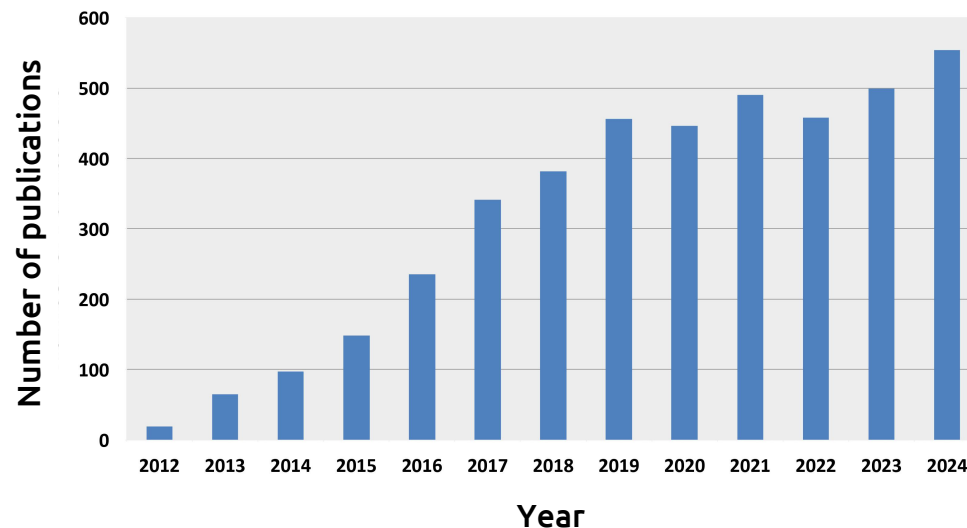


EH scale resolution

Main science categories

- Cosmology and the High-z Universe
- Galaxies and Galactic Nuclei
- ISM, Star Formation and Astrochemistry
- Circumstellar Disks, Exoplanets and the Solar System
- Stellar Evolution and the Sun

An incredibly productive facility for the community

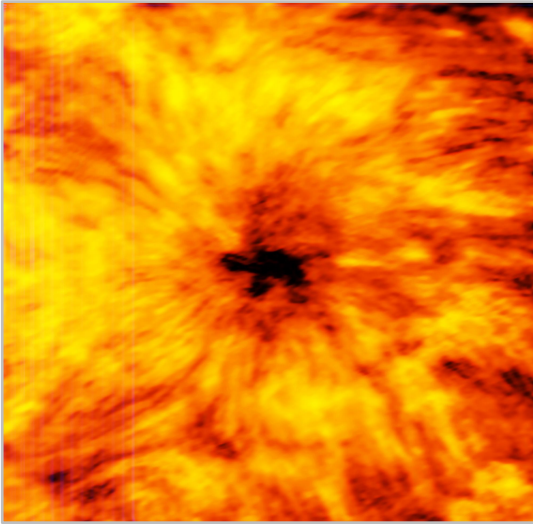


Credit: JAO

Our Solar System seen by ALMA

Sunspot at 1.25 mm

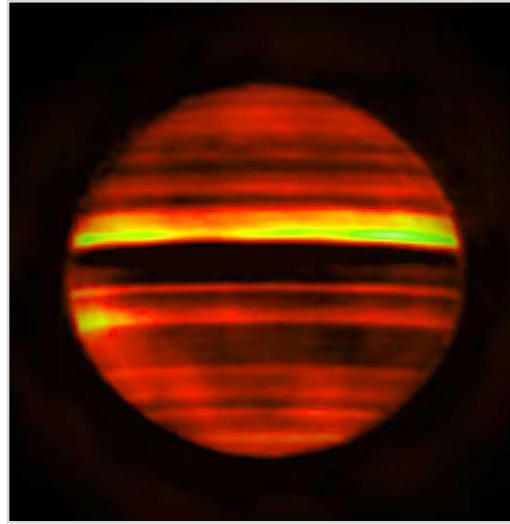
ESO/NAOJ/NRAO



240 GHz (Band 6)

Jupiter at 1.3 mm

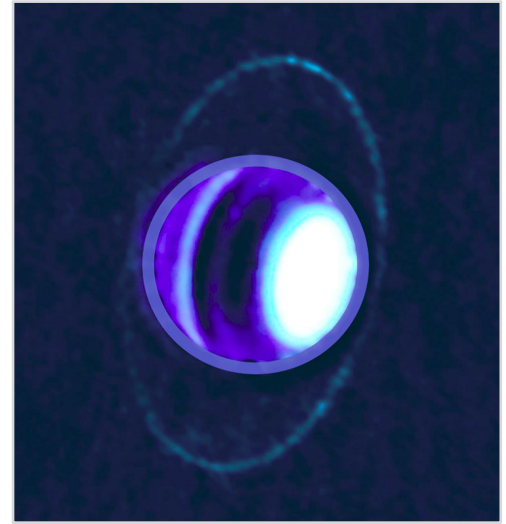
Pater et al. (2019)



230 GHz (Band 6)

Uranus at 1.3–3.1 mm

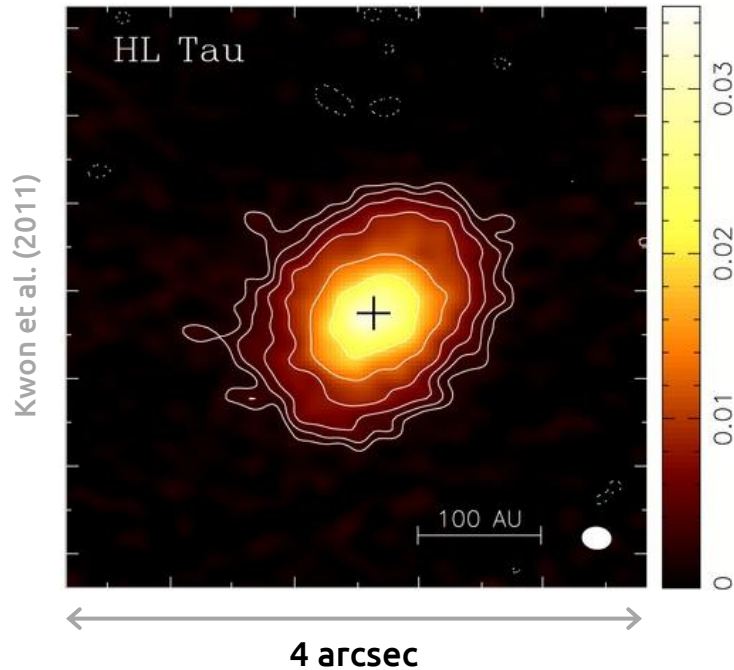
Molter et al. (2021)



97–230 GHz (Bands 3, 4, 6)

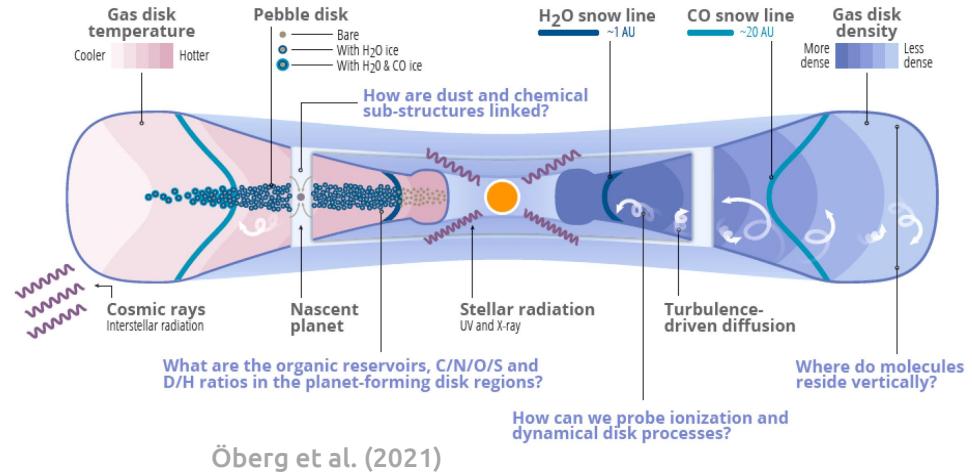
BEFORE ALMA

HL Tau protoplanetary disk



Protoplanetary disk studies

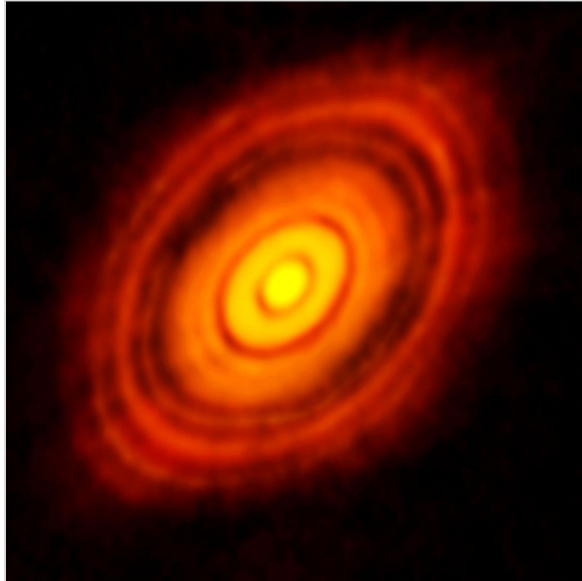
- Most young stars (~ 1 Myr) are surrounded by rotating disks of gas and dust $\rightarrow$ sites of planet formation
- Disk mass, size, composition $\rightarrow$ key factors that set the properties of newly formed planetary systems



THE ALMA REVOLUTION

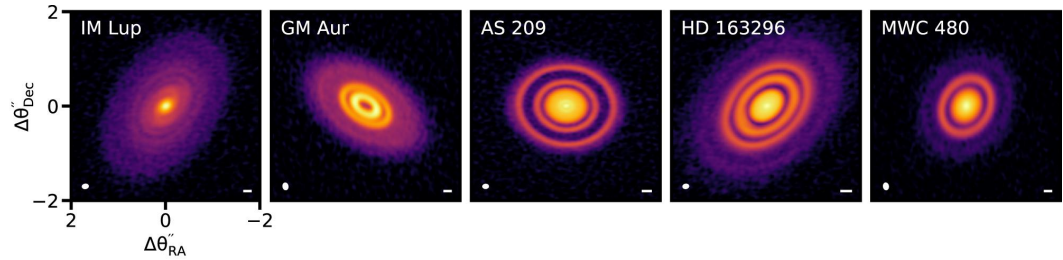
HL Tau protoplanetary disk

ALMA Partnership et al. (2015)

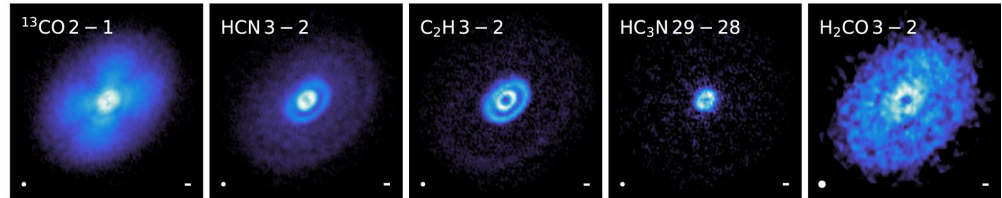


1.4 arcsec

ALMA dust continuum morphologies at 220 GHz



Molecular emission morphologies in HD 163296

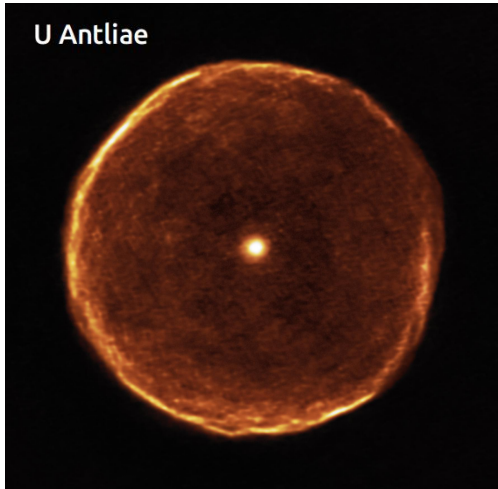


Disk chemical composition, structure, kinematics

Öberg, Guzmán et al. (2021)

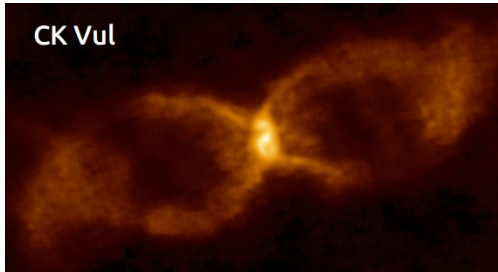
ALMA images of stars

F. Kerschbaum



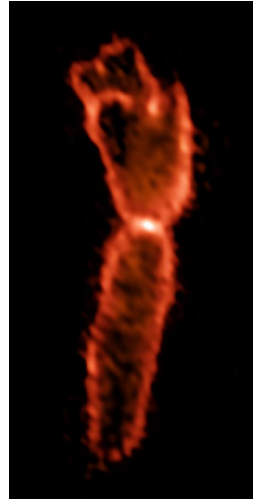
- Material ejected by the star: expanding circumstellar shell

S.P.S. Eyres



- ▲ Dust from a stellar collision

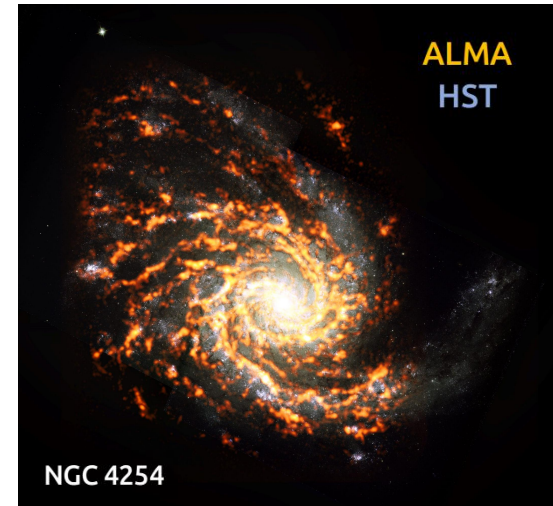
R. Sahai



- ▲ Pre-planetary nebula produced by a dying star

Stellar recycling: chemical enrichment

- Stars synthesize metals and molecules
- Late life stages → stellar mass loss
- The ISM becomes chemically enriched
- New clouds inherit this material

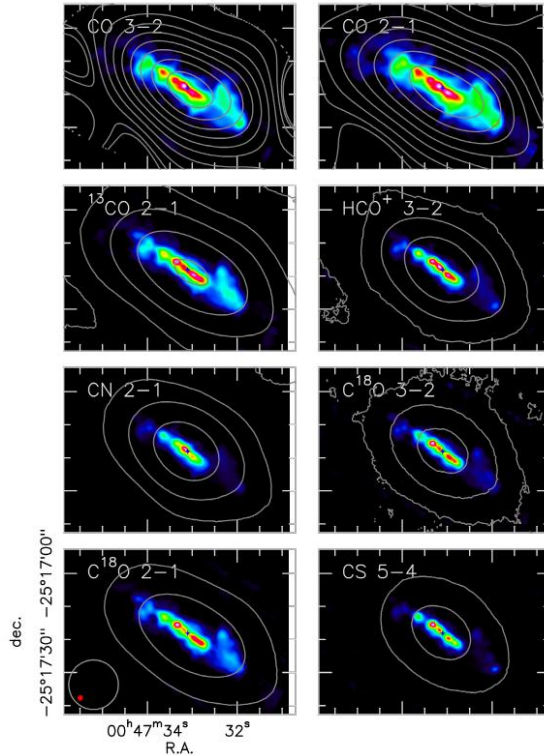


S. Dagnello (NRAO)

ALMA PHANGS

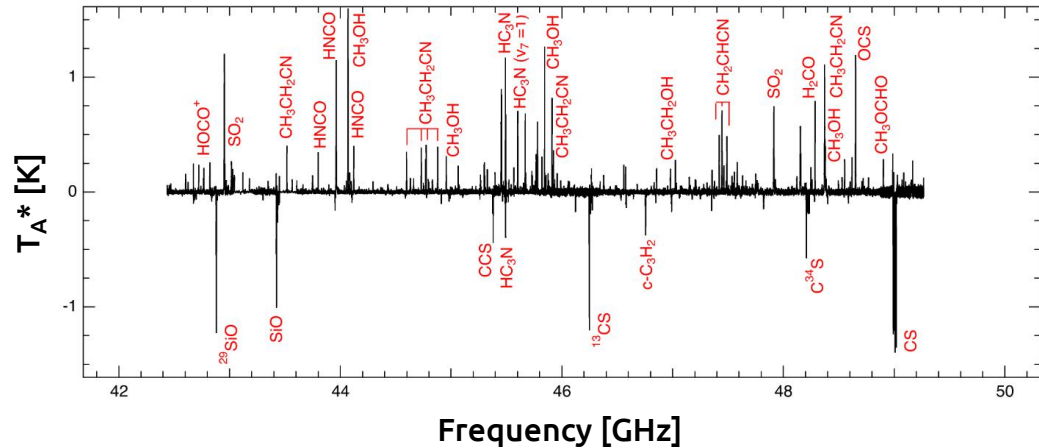
Mapping organic molecules → chemistry, physical conditions, earliest stages of star formation

Nearby galaxy NGC 253



Extremely rich molecular spectrum of a Milky Way cloud

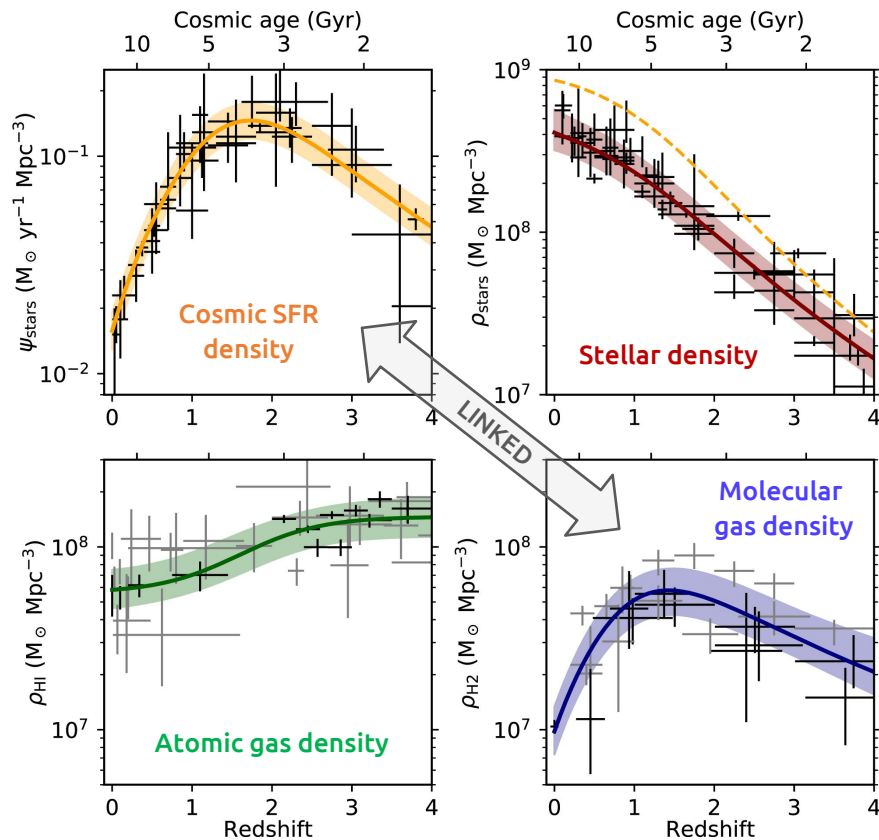
Sagittarius B2 cloud, Credit: Brett McGuire (MIT/NRAO)



Martín et al. (2021)

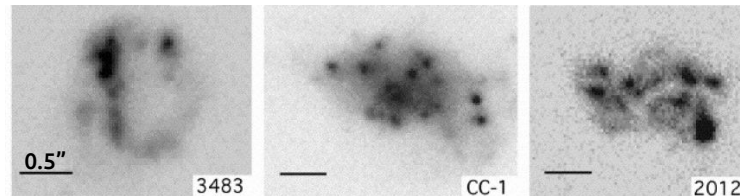
ALCHEMI Survey

CO, HCO⁺, CN, CS...

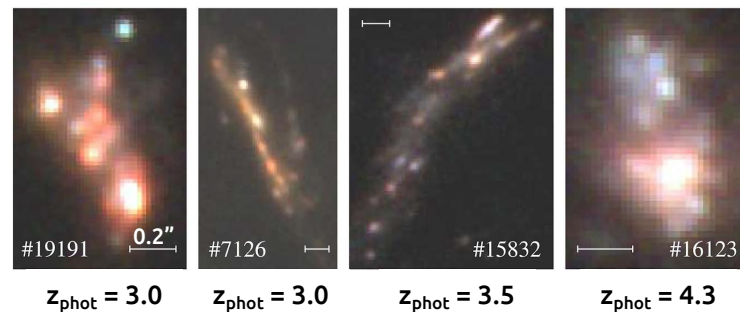


Adapted from Walter et al. (2020)

At cosmic noon: turbulent disks, stellar clumps



HST, $1.5 \leq z \leq 3$, Elmegreen et al. (2005)

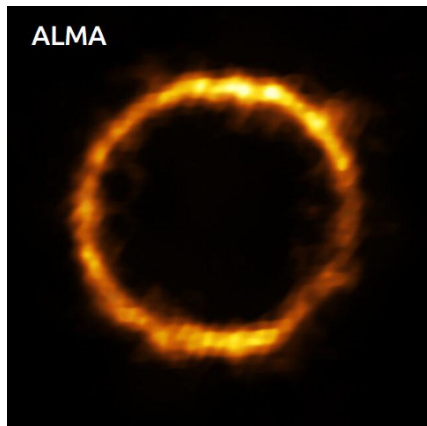


JWST/NIRCam, Claeysens et al. (2025)

To understand how bright rest-frame UV stellar clumps form, we need to resolve their parent GMCs $\rightarrow$ sub-kpc CO/[CII] obs.

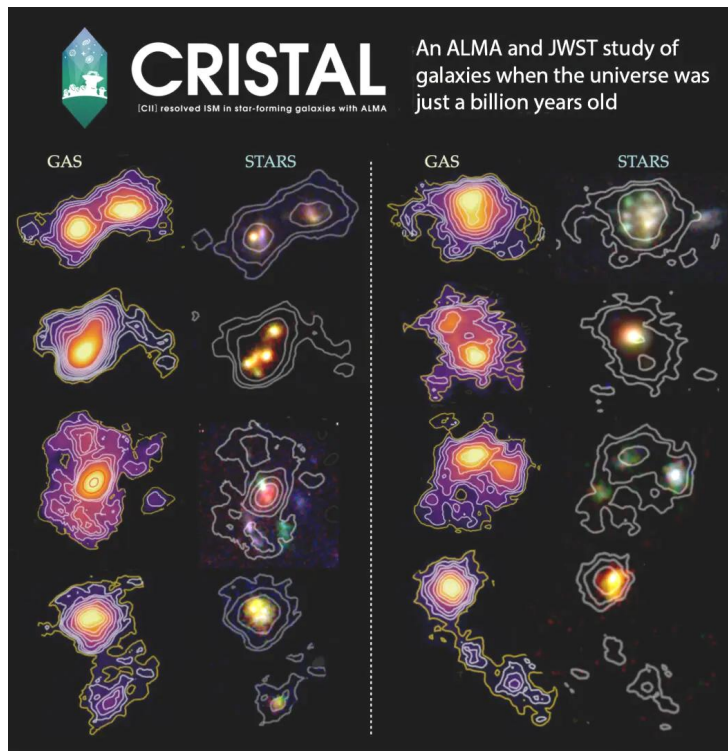
Resolving high- z galaxies with ALMA and JWST

Einstein ring at $z = 4.2$ (SPT0418-47)



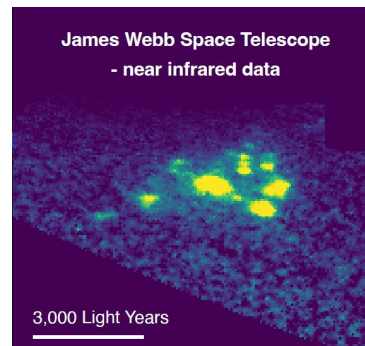
Rizzo et al. (2020)
Nature

Interplay between gas and stars at $4 < z < 6$

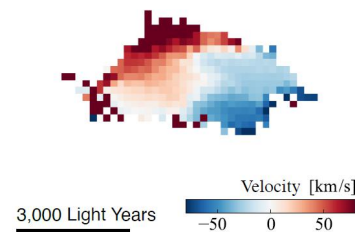


Herrera-Camus et al. (2025)

"Cosmic grapes" at $z \sim 6$



Atacama Large Millimeter Array (ALMA)
- Radio Data



Fujimoto et al. (2025)
Nature Astronomy

Examples of Current Limitations

Origins of Planets

High ($< 0.1 \text{ km s}^{-1}$) spectral resolution required $\rightarrow$ limited correlated bandwidth

Chemical Complexity

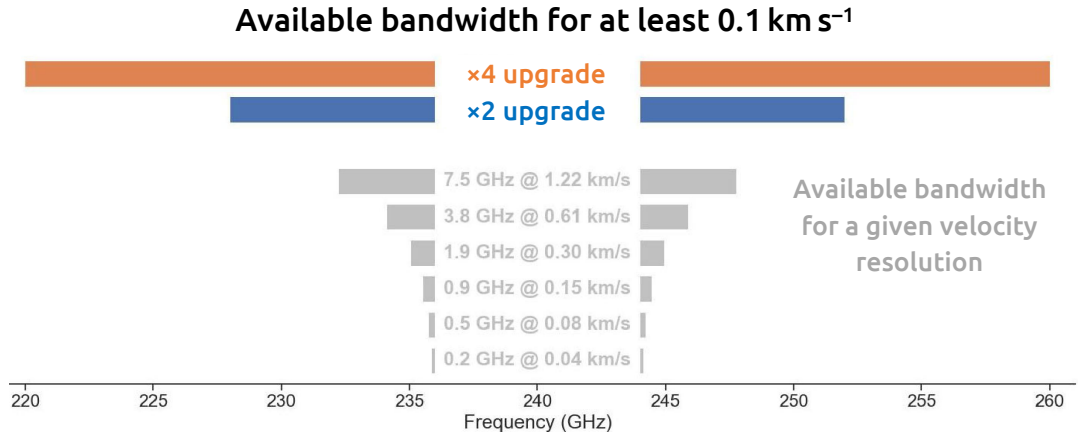
Limited number of lines per spectral scan $\rightarrow$ multiple scans required

Origins of Galaxies

Few lines in many galaxies (PHANGS) or many lines in one galaxy (ALCHEMI)

ALMA priorities toward 2040

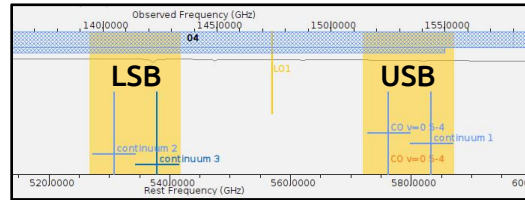
- 1) Increased receiver bandwidth
- 2) Optimized Science Archive
- 3) Extended maximum baseline
- 4) Higher number of 12m antennas



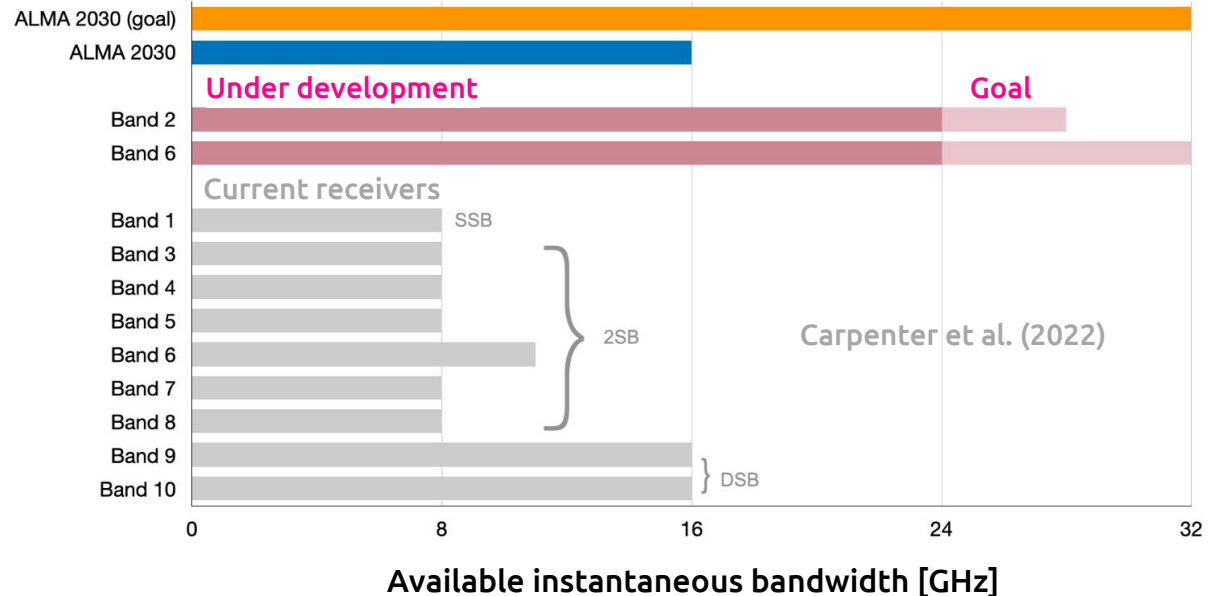
Current vs future correlated bandwidth in Band 6

The Wideband Sensitivity Upgrade (WSU)

- Increased bandwidth
 - 8×2 GHz/polarization
 - Enhanced sensitivity
 - Correlated bandwidth 0.1 km s^{-1} : $\times 10$ to $\times 100$
- Observing speed gains
 - Cont. imaging: $\times 3$ to $\times 6$
 - Line imaging: $\times 2$ to $\times 3$
 - Faster spectral scans, fewer frequency tunings
- Electronics upgrade
 - New correlator (OSF)
 - Better digital efficiency, improved sensitivity $\times 1.2$



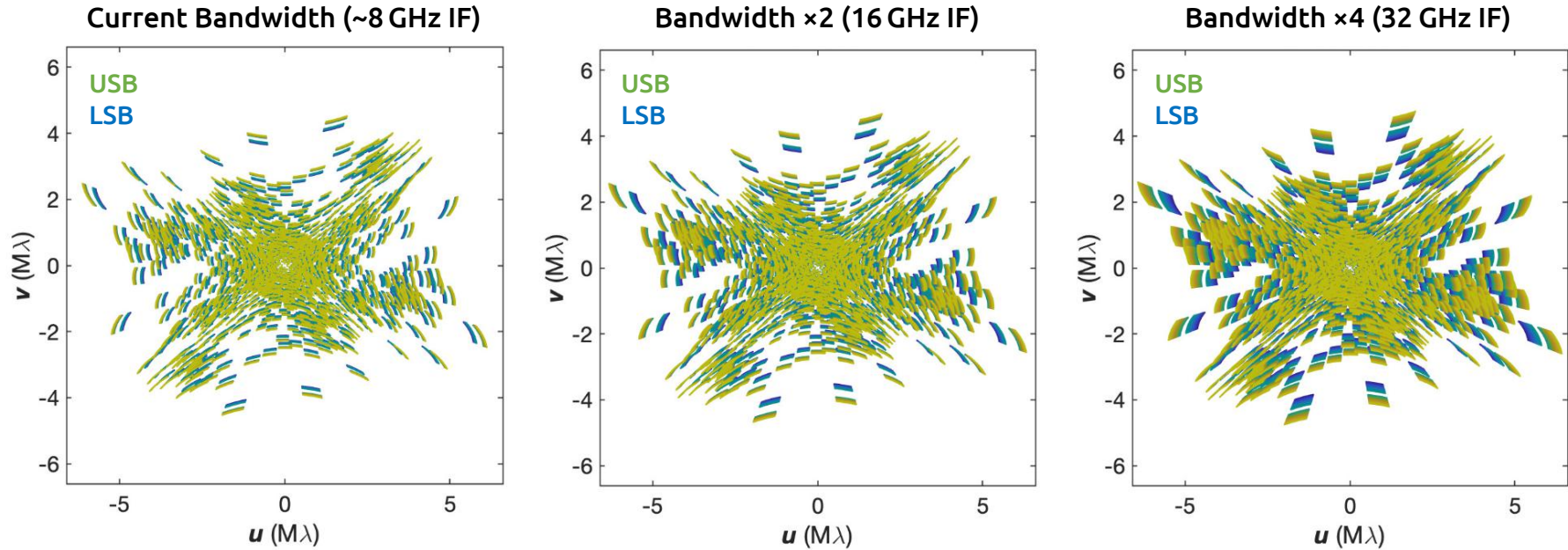
- Bandwidth $\times 2$, later $\times 4$
Current: $4 \times 1.875 = 7.5 \text{ GHz}$
2030 Goal: $8 \rightarrow 16 \rightarrow 32 \text{ GHz}$



The Wideband Sensitivity Upgrade (WSU)

Beyond sensitivity, the WSU also changes how ALMA samples the uv-plane.

- Denser uv-plane sampling → smaller USB/LSB gap → faster imaging, improved image reconstruction



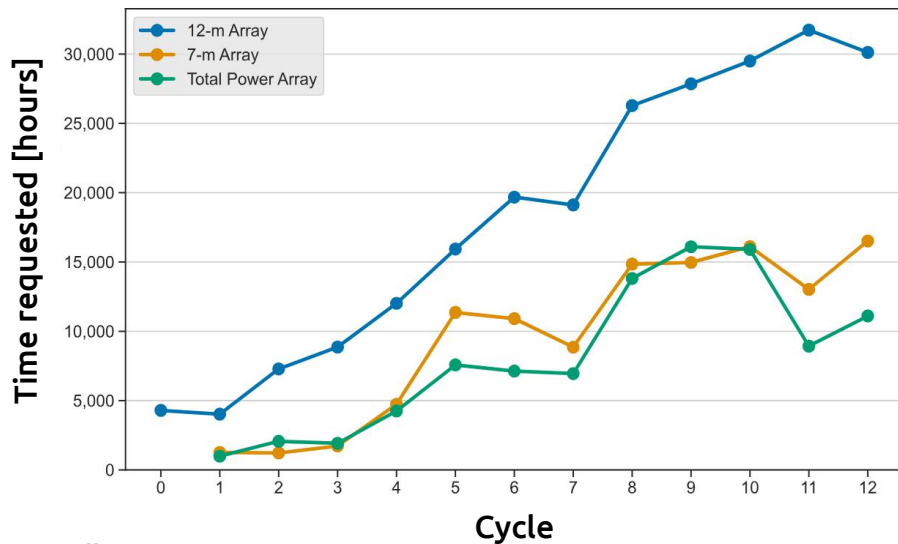
Adapted from Carpenter et al. (2022)



Practical Aspects & Training

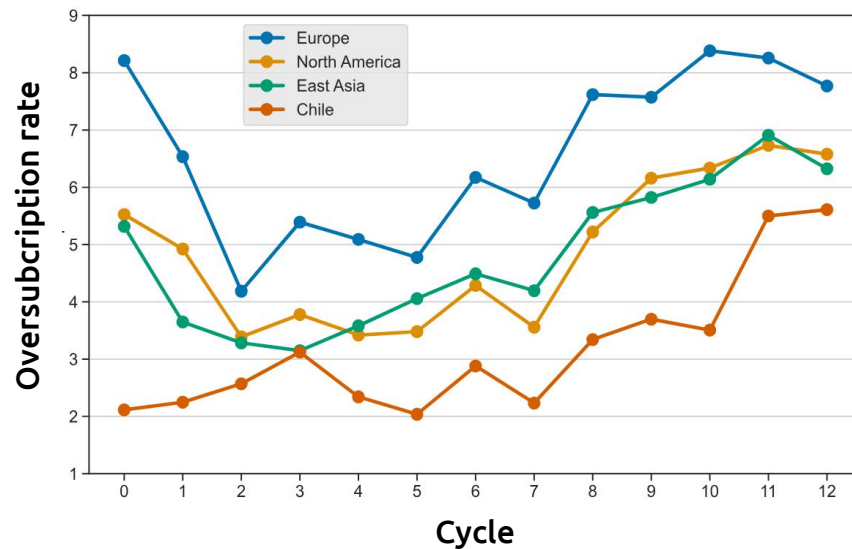
ALMA Observing Cycle Statistics

Time requested on the different arrays



Credit: JAO

Oversubscription rate on the 12-m Array



ALMA is highly competitive
Oversubscription rates ~6–8

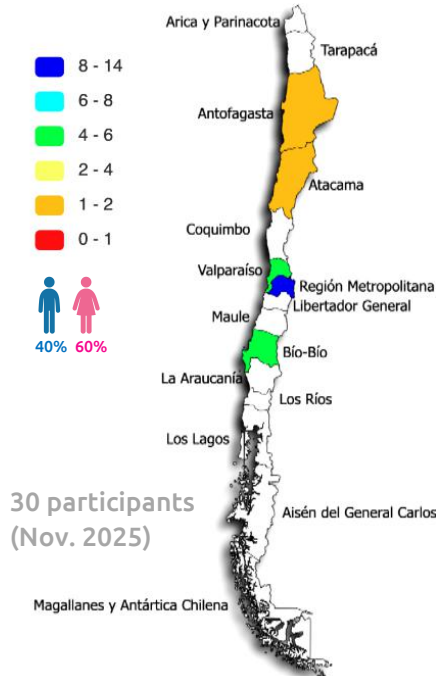


Local training opportunities

- **ALMA–NRAO workshops**
 - Proposal preparation
 - ALMA data reduction
- **Chilean ALMA Users School**
 - Classes & hands-on sessions
 - Visit of ALMA (OSF + AOS)



First Chilean ALMA Users School



Career opportunities

- **Student research visits**
 - JAO Visitor Program
 - ESO Studentship Program
- **Postdoctoral fellowships**
 - ESO–ALMA (Germany/Chile)
 - JAO Fellowship (Chile)
 - NRAO Jansky Fellowship (US)
 - Fondo ALMA–ANID (Chile)
- **ALMA Ambassadors Program**
 - Leading ALMA workshops
 - Funded travel to NRAO/NAASC
 - Research travel award



National Radio
Astronomy
Observatory

Socorro, New Mexico — 26 May–3 June 2026

Including a visit to the VLA and MROI sites

- **Week-long training for graduate students on aperture synthesis and interferometric imaging**
- **Hands-on tutorials on data collection, calibration, and imaging using VLA, VLBA, and ALMA observations**
- **Lectures on optical interferometry and visit to the Magdalena Ridge Observatory Interferometer (MROI)**
- **School website and registration link:**

<https://whova.com/web/Hs49eW8Z%40XxATfbPP3sKYyEAUFCmcHPF8SyHad%40doZo%3D/>